

ABSTRACT ALGEBRA 1 -PROBLEM SET - M. VICARÍA

Dear student please submit the complete solution to each of the following problems in Gradescope in a PDF file. Please write down your name in each page and specify clearly the problem that you stated. For your study purposes, I included a classification of the problems and their difficulties:

- Warm-up: these are require that you understand some definition.
- *: they indicate some medium level of difficulty. You should spend longer in these problems.
- ** indicates that the problem is challenging, they will require more time than the other problems.

The complexity from ** and * is the level we would like you to master in this course. Questions highlighted with **OPTIONAL** are not necessary to get full credit for the homework, but I highly encourage you to solve them for your learning process.

In each of the solutions you **SHOULD JUSTIFY COMPLETELY YOUR ANSWER** with a proof or a counter-example. No credit is granted to solutions without justification.

1. MANDATORY PROBLEMS

Definitions:

- (1) Let G be a group. The center of G is defined as $Z(G) = \{h \in G \mid \forall g \in G \ h \cdot g = gh\}$.
- (2) Let G be a group, we say that it is solvable if there is a finite sequence of subgroups $e = H_0 \leq H_1 \leq \dots H_{n-1} \leq H_n = G$ such that:
 - for every $1 \leq i \leq n-1$ H_i is normal in H_{i+1} ,
 - the quotient group H_{i+1}/H_i is abelian.
- (1) Every group of prime power is solvable.
- (2) The center of a finite non-trivial p - group is non trivial.
- (3) If p and q are distinct primes and $p < q$ then every group G of order pq has a single subgroup of order q and this subgroup is normal in G . Hence G is not simple. If we further assume that q is not congruent with 1 modulo p then G is abelian and cyclic.
- (4) No group of order 30 is simple.
- (5) No group of order 48 is simple.
- (6) No group of order 36 is simple.
- (7) Determine whether the following are true or false:
 - (a) Every group of order 255 is cyclic, therefore abelian.
 - (b) Every group of order 159 is cyclic.
 - (c) Every group of 125 elements has at least 5 that commute with every element of the group.